

GLARUS, Switzerland

WMO: 066850

Lat: 47.0347N

Lon: 9.0669E

Elev: 1698

StdP: 13.82

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.3	17.7	7.7	8.7	17.5	12.0	10.8	19.5	27.7	47.4	23.0	39.7	2.2	140	0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	85.5	67.8	81.9	66.3	78.7	65.1	69.4	81.8	67.8	79.2	66.3	76.7	11.2	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.1	99.0	75.5	63.6	93.8	73.2	62.4	89.6	71.4	34.5	81.9	33.1	79.3	31.9	76.8	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.4	20.1	16.7	DB	11.3	90.8	4.1	2.8	8.3	92.9	5.9	94.5	3.5	96.1	0.5	98.2	
			WB	10.0	73.2	4.3	1.6	6.9	74.4	4.3	75.3	1.9	76.2	-1.3	77.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	32.1	33.9	41.7	50.3	56.5	63.8	66.3	65.0	57.8	50.9	41.3	33.9	(6)
	DBStd	13.93	7.64	7.62	7.35	7.13	6.87	6.66	5.86	5.88	5.80	6.61	7.32	7.10	(7)
	HDD50	2229	555	450	266	83	20	1	0	0	7	71	275	501	(8)
	HDD65	5891	1020	870	722	441	273	101	56	74	222	436	712	964	(9)
	CDD50	2065	0	0	9	92	222	415	504	466	242	100	13	2	(10)
	CDD65	254	0	0	0	0	10	65	96	75	8	0	0	0	(11)
	CDH74	2012	0	0	0	12	129	561	740	520	45	5	0	0	(12)
	CDH80	511	0	0	0	0	16	145	211	137	2	0	0	0	(13)
(14) Wind	WSAvg	4.8	3.5	4.1	5.2	6.6	6.3	5.8	5.3	4.8	4.2	3.8	3.9	3.4	(14)
(15) Precipitation	PrecAvg	72.0	4.3	4.6	5.2	4.9	7.0	7.9	8.5	8.3	6.1	4.8	5.2	5.2	(15)
	PrecMax	93.1	9.8	13.9	10.3	9.6	14.5	11.9	14.4	14.5	11.4	9.8	13.5	12.7	(16)
	PrecMin	55.4	0.3	1.4	2.0	1.1	1.9	4.0	3.3	2.0	2.6	0.4	0.0	0.3	(17)
	PrecStd	8.1	2.6	2.8	2.5	2.0	3.1	2.2	3.0	3.0	2.1	2.2	2.9	2.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.8	57.1	65.4	75.6	82.0	88.1	90.4	89.4	78.6	74.2	67.9	62.7	(19)
		MCWB	43.1	44.5	49.3	56.4	62.2	69.1	68.6	69.0	64.0	56.2	50.8	47.4	(20)
	2%	DB	47.5	51.6	61.3	71.1	77.5	84.2	85.6	83.6	74.7	67.4	62.8	49.4	(21)
		MCWB	42.7	41.9	48.0	54.3	60.4	67.7	67.5	68.2	63.1	56.4	48.5	43.8	(22)
	5%	DB	44.1	47.0	58.0	67.8	73.7	80.6	81.6	79.8	70.8	63.5	54.7	45.1	(23)
		MCWB	40.5	40.2	46.8	52.9	58.7	65.5	66.4	66.8	61.6	55.6	48.2	42.1	(24)
	10%	DB	41.3	43.4	53.7	63.8	69.7	76.6	78.0	75.9	67.5	60.4	50.6	42.7	(25)
		MCWB	38.4	38.8	45.0	51.0	57.3	64.0	65.3	65.3	60.1	54.6	46.7	40.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.4	46.4	51.5	58.2	65.3	71.8	71.7	71.5	67.0	60.2	52.2	47.7	(27)
		MCDB	49.6	51.9	60.6	72.5	79.2	85.7	85.0	83.7	75.4	65.5	60.3	58.8	(28)
	2%	WB	42.9	43.3	49.4	55.8	62.4	68.5	69.1	69.2	64.4	58.1	50.4	44.7	(29)
		MCDB	46.3	49.9	58.8	68.8	74.2	81.1	81.6	81.0	72.3	63.6	58.5	48.7	(30)
	5%	WB	40.7	41.0	47.5	53.8	60.1	66.7	67.5	67.4	62.4	56.5	48.8	42.5	(31)
		MCDB	44.1	46.3	56.0	65.5	70.1	78.2	78.9	77.9	69.5	62.0	55.2	45.1	(32)
	10%	WB	38.6	38.8	45.5	51.7	58.1	64.9	66.0	65.8	60.6	55.0	46.9	40.2	(33)
		MCDB	41.3	43.0	52.8	61.8	67.8	75.0	76.4	74.7	66.2	59.7	51.0	42.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.7	11.5	15.2	17.8	17.7	17.6	17.5	16.3	14.4	13.7	11.0	9.4	(35)
		MCDBR	13.1	17.4	22.3	24.9	25.4	24.7	24.5	22.9	19.6	18.3	16.7	13.0	(36)
	5% WB	MCWBR	8.6	10.3	12.4	12.6	12.6	11.9	10.6	11.1	10.5	10.0	9.0	8.4	(37)
		MCWBR	12.6	15.7	19.9	22.7	22.2	22.4	21.8	20.6	17.9	15.9	14.6	11.8	(38)
(40) Clear-Sky Solar Irradiance	taub		0.279	0.308	0.355	0.401	0.420	0.428	0.428	0.418	0.385	0.352	0.312	0.277	(40)
		taud	2.337	2.268	2.228	2.199	2.216	2.245	2.246	2.289	2.363	2.427	2.450	2.385	(41)
	Ebn at Noon		257	271	273	270	269	266	264	262	261	253	243	244	(42)
		Edh at Noon	23	31	37	43	43	43	42	39	33	26	21	20	(43)
(44) All-Sky Solar Radiation	RadAvg		431	716	1100	1393	1552	1800	1766	1538	1173	757	444	357	(44)
	RadStd		50	90	153	212	156	165	158	149	118	85	60	50	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (122 neighbors)	N/A	N/A	N/A	N/A	+1.21	+1.28	N/A	N/A	N/A	N/A
		N/A	N/A	N/A	+1.24	+0.72	+0.58	N/A	N/A	+103	+27

Nomenclature: See separate page